

Update to the Neonatal Early-Onset Sepsis Calculator Utilizing a Contemporary Cohort

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BACKGROUND AND OBJECTIVES: The Kaiser Permanente Neonatal Early-Onset Sepsis (EOS) Calculator has been an effective tool for risk stratification to safely reduce newborn antibiotic exposure. The calculator was derived from data on infants born between 1993 and 2007. Since that time, US obstetric practice has adopted universal antepartum screening for group B *Streptococcus* and intrapartum antibiotic prophylaxis guidance has changed. Our objective was to update the EOS calculator using a contemporary birth cohort and determine the effect of these changes on EOS case ascertainment and antibiotic recommendations.

METHODS: The study included infants born at ≥ 35 weeks' gestation at 14 hospitals between January 2010 and December 2020 ($n = 412\,595$ infants, EOS cases = 113). Model coefficients were re-estimated and the point estimates of the likelihood ratios for clinical status used to calculate the posterior probability of EOS. We compared the number of EOS cases correctly identified by each model (sensitivity) and the proportion of infants for whom empirical antibiotics are recommended.

RESULTS: The original model had a sensitivity of 0.76 (95% confidence interval 0.63–0.85), while the updated model had a sensitivity of 0.80 (95% confidence interval 0.68–0.89), $P = .15$. The recommended empirical antibiotic use was 3.5% with the original model and 3.7% with the updated model, $P < .0001$. For each additional case identified by the updated model, an additional 158 infants would be treated with antibiotics.

CONCLUSIONS: Both the original and updated EOS calculators are effective tools for quantifying EOS risk among infants born at ≥ 35 weeks' gestation.

abstract



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WHAT'S KNOWN ON THIS SUBJECT: The Neonatal EOS Calculator has been an effective tool for risk stratification to safely reduce newborn antibiotic exposure. Since its development, US obstetric practice has adopted universal group B *Streptococcus* antepartum screening, and intrapartum antibiotic prophylaxis guidance has changed.

WHAT THIS STUDY ADDS: An updated calculator using a contemporary cohort provides equivalent sensitivity in identifying EOS cases compared with the original calculator, with a small increase in empirical antibiotic use. Both are effective risk stratification tools for quantifying the risk of EOS.

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In 2011, Puopolo et al developed a risk prediction model to estimate the probability of neonatal early-onset sepsis (EOS) on the basis of maternal risk factors known at the moment of birth.¹ Risk estimation was improved with a model incorporating infants' clinical condition as it evolved over the first 24 hours after birth.² These models were combined with clinical care recommendations in what is now known as the Kaiser Permanente Neonatal EOS Calculator (KP EOS Calculator).^{3,4} The implementation of this approach in Kaiser Permanente Northern California (KPNC) birth centers resulted in a 66% reduction in blood cultures and a 48% reduction in antibiotic use in the first 24 hours after birth.⁴ Safety was not compromised; we observed no change in antibiotic initiation after 24 hours, severe neonatal illness, or readmission for EOS. The application of the KP EOS calculator in US and international populations has confirmed its effectiveness in reducing newborn antibiotic use.⁵⁻⁹ In 2018, the American Academy of Pediatrics endorsed the calculator as one approach to evaluate a newborn's risk of EOS.¹⁰

Although effective, does the model need updating? The calculator models were derived from data on infants born between 1993 and 2007. Since 1993, US obstetric practice has adopted universal antepartum screening for group B *Streptococcus* (GBS), and guidance has changed in regard to which antibiotics constitute appropriate intrapartum antibiotic prophylaxis (IAP).¹¹⁻¹⁵ The original models included infants born at 34 weeks' gestation; the high prevalence of respiratory distress among such infants diminishes the predictive value of their clinical status in EOS prediction. In most cases, decisions regarding EOS risk are not the primary reason these late preterm infants are admitted to intensive care units. In addition, some of the methodologic choices in the original calculator have been questioned.¹⁶ In addition, the original model used a nested case-control design that required statistical adjustment to the intercept for the model when applied to settings with a different baseline EOS population incidence.

Our objective was to update the KP EOS calculator using a contemporary birth cohort of infants born at ≥ 35 weeks' gestation and determine the effect of these changes on EOS case ascertainment and antibiotic recommendations.

METHODS

Population

The study population comprised infants born at ≥ 35 weeks' gestation at 14 KPNC hospitals between January 2010 and December 2020. Infants who died in the delivery room and those with major congenital anomalies were excluded. We compared the performance of the original model to the updated model using data from infants born from July 2014 to December 2020 (the period the original calculator was in clinical use at KPNC).

Original Prediction Model

The original multivariate model predicts the risk of EOS at birth using gestational age (GA; continuous), highest maternal temperature 24 hours before delivery (continuous), GBS carrier status (negative, positive, or unknown), the duration of the rupture of membranes (continuous), and intrapartum antibiotic treatment (none; GBS IAP given at any time before delivery or broad-spectrum antibiotics given < 4 hours before delivery and broad-spectrum antibiotics given ≥ 4 hours before delivery).¹ A second model incorporates clinical condition (well-appearing, equivocal status, or clinical illness).² The risk estimate at birth from the first model (the modified prior probability of EOS) is then multiplied by the upper limit of the 95% confidence limit of the likelihood ratio (LR) for the corresponding clinical status in the second model as it evolved over the first 24 hours after birth to obtain the final posterior probability of EOS.

Data Sources and Definitions

From existing KPNC databases, we obtained maternal race and ethnicity, GA, the duration of the rupture of membranes, and the mode of delivery.^{17,18} We extracted the highest maternal temperature before delivery from inpatient vital sign flowsheets; 98.6 °F (37 °C) was assigned if no temperature was recorded. We abstracted maternal antibiotic administration before delivery from the electronic medication administration record and maternal GBS status and infant blood culture results from laboratory databases. To determine clinical status, we employed laboratory results, medication data, respiratory care flowsheets, and nursing flowsheets. Clinical status was determined at birth, at 12 hours, and at 24 hours. Infants were considered to have clinical illness if (1) the infant received supplemental oxygen, nasal continuous positive airway pressure, or high-flow nasal cannula (> 1 L/min) at birth and for at least 2 hours after birth, (2) the 5-minute Apgar score was < 5 , (3) the cord blood or infant blood gas at < 2 hours after birth had a base deficit of ≥ 16 , (4) vasoactive drugs (dopamine, dobutamine, epinephrine, or vasopressin) were administered, or (5) the infant required mechanical ventilation. If respiratory support was started after 1 hour, the infant was considered clinically ill when respiratory support was started. The infant status was classified as equivocal if there was a persistent physiologic abnormality, as follows: heart rate ≥ 160 , respiratory rate ≥ 60 , temperature ≥ 100.4 °F (38 °C), or temperature < 97.5 °F (36.4 °C) on 2 occasions separated by at least 4 hours. Additionally, 2 abnormalities on 2 occasions, separated by 2 hours qualified as equivocal. Infants were considered well-appearing if they did not have a clinical illness or equivocal status.

Revisions to the Model

Classification of Maternal Antibiotics

The original prediction model classified erythromycin, clindamycin, and vancomycin as GBS IAP per CDC 2002 recommendations.¹⁹ With changes in GBS antibiotic susceptibility and updated information on transplacental antibiotic transfer, revisions were made to GBS IAP recommendations.^{11–15,20–23} For the updated model, only the administration of ampicillin, penicillin, and cefazolin was considered adequate for GBS IAP.^{11,23} We classified GBS IAP in combination with an aminoglycoside, aztreonam, azithromycin, or metronidazole as broad-spectrum antibiotics. Also included as broad-spectrum antibiotics were second-, third-, or fourth-generation cephalosporins, extended-spectrum penicillins, fluoroquinolones, carbapenems, tetracyclines, and sulfamethoxazole/trimethoprim.

The timing of antibiotics was classified in the original model as antibiotics given ≥ 4 hours before delivery, antibiotics given < 4 hours before delivery, and no antibiotics given.¹ For clinical implementation, antibiotics given < 2 hours before delivery were classified as “no antibiotics.” GBS IAP administered ≥ 2 hours before delivery was considered adequate IAP because the effectiveness of IAP in preventing GBS-specific EOS, if given ≥ 2 hours, is $\geq 89\%$.^{24–27}

For the updated model, we classified antibiotic timing as ≥ 4 hours before delivery, 2 to 3.9 hours before delivery, < 2 hours before delivery, or no antibiotics.

Recalibration of Model Coefficients

Using the entire cohort of infants born at ≥ 35 weeks' gestation between January 2010 and December 2020, we estimated the coefficients of the updated model using logistic regression. EOS was defined as isolation of a bacterial pathogen from a blood culture obtained ≤ 72 hours after birth. Cultures growing commensal organisms (eg, coagulase-negative staphylococci) were classified as EOS if the clinical team considered the infant infected, as evidenced by an antibiotic treatment of ≥ 5 days or until neonatal death. Model performance was evaluated using bootstrap resampling to estimate the confidence interval (CI) for the area under the receiver operating curve and calculate the integrated calibration index, which is equivalent to the mean difference between predicted probabilities and observed probabilities derived from a smoothed calibration curve.^{28–30}

Likelihood Ratios for Clinical Status

In the original calculator, the upper limit of the 95% CI of the LR for the clinical status was used to calculate the EOS posterior probabilities.^{3,4} This was done intentionally to estimate the highest risk, as an added measure of safety. Now that numerous implementation studies have revealed the safety of this approach,^{5–9} in the updated

model, we used the point estimates of the LRs to eliminate bias in the risk estimates. The LRs used were clinical illness, 14.5 (original LR 21.2), equivocal, 3.65 (original 5.0), and well-appearing, 0.36 (original 0.41).²

Intercept Estimation and Adjustment

The original model was developed using all EOS cases and a random sample of controls, resulting in a higher outcome prevalence than in the base population. This required an adjustment to the model intercept to reflect the population proportion of case subjects and controls. This bootstrap-based method for intercept adjustment is compared with the standard method in Supplemental Information.^{16,31} The intercept of the revised model was calculated directly using the entire birth cohort. In applying the model to populations outside of KPNC, it may be necessary to adjust the intercept for the prevalence of EOS in that population if the difference is due to factors other than differences in the prevalence of the predictors in the model. Supplemental Information details the methodology for adjustment.

Comparison of Models

The study population born July 2014 through December 2020 was used to compare the models. We compared the overall distribution of EOS risk scores at birth among the study population, the number of EOS cases correctly identified by each model (sensitivity), and the proportion of study infants for whom empirical antibiotics are recommended. Empirical antibiotics are recommended if the infant has an EOS risk score of ≥ 3 cases per 1000 live births or becomes clinically ill. Sensitivities were compared using McNemar's test with Yates correction.

The KPNC Institutional Review Board approved this data-only study and waived informed consent.

RESULTS

The study cohort included 412 543 infants born between January 2010 and December 2020. During this period, 113 cases of culture-confirmed EOS were identified (incidence 0.27 cases of EOS per 1000 live births). Population characteristics are provided in Table 1. Approximately 5% of the cohort was born at < 37 weeks' gestation; GBS screening results were known in 95.5% of the cohort. In the first 24 hours after birth, 89% of the cohort met the criteria for well-appearing clinical status. At birth, 1.5% of infants met the criteria for clinical illness, with an additional 1.4% becoming clinically ill in the first 24 hours. Among EOS cases, 50 (44%) had signs of illness at birth, with an additional 33 (29%) exhibiting signs by 24 hours after birth; only 6 (5%) developed signs ≥ 48 hours. No signs of illness were identified in 18 (16%) EOS cases, and in 8 of 18, a repeat blood culture was sterile before antibiotics were administered. The characteristics of all

TABLE 1 Population Characteristics		
	EOS Cases	Population
<i>n</i>	113	412 543
Maternal age, mean (SD)	29.1 (5.7)	30.8 (5.5)
Maternal race/ethnicity, <i>n</i> (%)		91 539 (22.2)
Asian	25 (22.1)	91 539 (22.2)
Black	10 (8.8)	27 077 (6.6)
Hispanic	27 (23.9)	94 775 (23.0)
White	37 (32.7)	148 969 (36.1)
Other	12 (10.6)	49 025 (11.9)
Multiple gestation, <i>n</i> (%)	5 (4.4)	10 983 (2.7)
Cesarean delivery, <i>n</i> (%)	34 (30.1)	107 969 (26.2)
GA, <i>n</i> (%)		
35–36 wk	11 (9.7)	21 301 (5.2)
37–40 wk	78 (69.0)	344 611 (83.5)
≥41 wk	22 (19.5)	46 629 (11.3)
GBS status, <i>n</i> (%)		
Positive	15 (13.3)	87 379 (21.2)
Negative	87 (77.0)	306 528 (74.3)
Unknown	9 (8.0)	18 634 (4.5)
Intrapartum maternal antibiotics, <i>n</i> (%)		
GBS-specific	9 (8.0)	75 033 (18.2)
Broad-spectrum	5 (4.4)	14 764 (3.6)
Maternal fever ≥100.4 °F (38.0 °C), <i>n</i> (%)	31 (27.4)	32 400 (7.9)

113 EOS cases are presented in Supplemental Tables 6 and 7.

β coefficients and odds ratios for the predictive model are noted in Table 2. The model provides adequate discrimination and calibration (Table 2).

Comparing the risk of EOS at birth among infants born July 2014 to December 2020 (*n* = 264 347) to those infants included in the original model, the overall distribution of

TABLE 2 Updated EOS Model Coefficients and Performance Statistics			
	β Coefficient	OR	P
Intercept	57.29929499		.22120
Highest maternal temperature ^a	0.85194656	2.344	.00000
GA ^b	−7.72247124	0.000	.00127
GA squared	0.09842383	1.103	.00141
Rupture of membranes ^c	0.86770862	2.381	.00016
Abx1 ^d	−2.13142945	0.119	.00000
Abx2 ^e	−2.33985917	0.096	.00012
GBS-positive	1.02265353	2.781	.00102
GBS Status Unknown	1.13710111	3.118	.00267
Bootstrapped performance metrics	Mean	95% CI	
AUC-ROC	0.788	0.737	.829
Integrated Calibration index	0.000086	0.000025	.000175

Abx, antibiotics; AUC-ROC, area under the curve, receiver-operator characteristic; OR, odds ratio.

^a Temperature is captured to nearest 0.1°F.

^b GA is captured in exact days.

^c Time in h + 0.05 to the 1/5th power.

^d Abx1 = 1 if GBS IAP given ≥2 h before delivery or broad-spectrum antibiotics given ≥2 h but <4 h before delivery.

^e Abx2 = 1 if broad-spectrum antibiotics given ≥4 h before delivery.

EOS risk for both models is right-skewed (Fig 1). The updated model is less skewed, with a higher number of infants with an EOS risk, as follows: ≥0.5 cases per 1000 live births.

Using the July 2014 to December 2020 cohort, we compared the sensitivity of the models after applying the newborn's clinical status at birth, 12 hours, and 24 hours (Table 3). There was no statistically significant difference in sensitivity at any time point, as follows: at birth, 0.45 (95% CI 0.33–0.58) versus 0.48 (95% CI 0.36–0.61), *P* = .29; by 12 hours, 0.65 (95% CI 0.52–0.76) versus 0.70 (95% CI 0.57–0.80), *P* = .15; and by 24 hours, 0.76 (95% CI 0.63–0.85) versus 0.80 (95% CI 0.68–0.89), *P* = .15. Comparing the original to the updated calculator, there was no difference in recommended antibiotic use at birth (1.7% vs 1.7%), a 0.1% increase by 12 hours (3.2% vs 3.3%), and a 0.2% increase by 24 hours (3.5% vs 3.7%). The updated calculator recommended empirical antibiotics (EOS risk ≥3 per 1000 live births) or a blood culture (EOS risk 1–2.9) in the first 24 hours in 88% (58/66) of EOS cases. For each additional case identified in the first 24 hours with the updated model, an additional 158 infants would be treated with empirical antibiotics (Table 3).

Table 4 details EOS cases in which either the original or updated model did not recommend the initiation of empirical antibiotics in the first 24 hours after birth. There were 3 EOS cases (cases 1, 2, 7) in which the updated EOS calculator recommended the initiation of empirical antibiotics, but the original calculator only recommended blood culture. In these cases, antibiotics were started after the culture was flagged positive; however, in all 3 cases, repeat blood cultures were sterile before the initiation of antibiotics. In 13 EOS cases, neither the original nor the updated calculator recommended the initiation of empirical antibiotics. In 2 of 13 (cases 3 and 4), the infants never exhibited signs of illness, and repeat blood cultures were sterile before the initiation of antibiotics. In 5 of 13 cases, the infants had equivocal signs within the first 24 hours, prompting a blood culture (cases 5, 6, 8–10); in cases 5 and 6, repeat blood cultures were sterile before the initiation of antibiotics. The 2 infants who developed signs of illness after 48 hours had less common EOS pathogens, *Staphylococcus aureus* and *Enterococcus faecalis*.

DISCUSSION

The updated KP EOS calculator exhibited similar performance in identifying EOS cases in the first 24 hours after birth as the original version. We will provide the updated version of the calculator for clinical use; however, our results suggest that either version can be used safely for EOS risk stratification. Although there was no statistical difference in the sensitivity of the 2 calculators, the updated model offered a slight advantage in case identification,

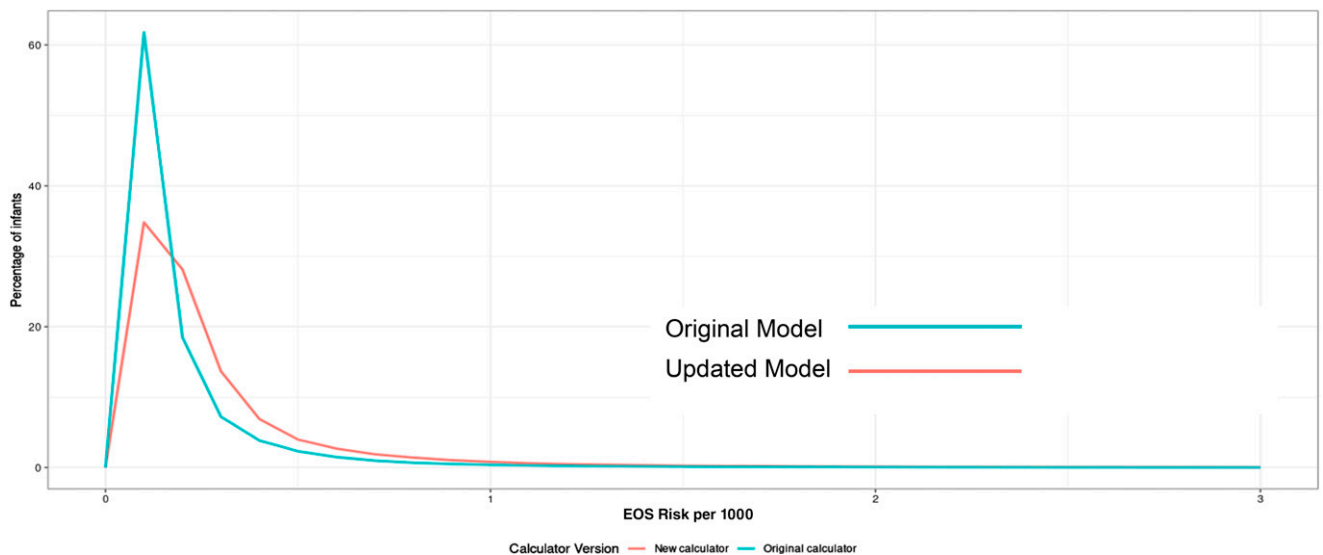


FIGURE 1
Distribution of EOS risk at birth by model.

balanced by a higher recommended use of empirical antibiotics in the first 24 hours after birth (3.7% vs 3.5%). Among 264 347 newborns, the updated calculator identified 1 additional EOS case in the first 24 hours at the expense of empirical antibiotics in an additional 158 uninfected infants.

Compared with the original model's coefficients,¹ the largest changes in coefficients in the updated model were those for GA (original model, GA -6.9325 , GA² 0.0877 ; updated model, GA -7.7225 , GA² 0.0984) and unknown GBS status (original model, GA 0.0427 ; updated model, 1.1371). Study population differences likely account for the differences in model coefficients. The original model included infants born at ≥ 34 weeks' gestation, whereas the updated model only includes infants born at ≥ 35 weeks' gestation. In the original study, GBS screening results were only available for 45% of the infants, whereas GBS results were known for 95.5% of the contemporary cohort.¹

Among the 66 EOS cases that occurred from July 2014 to December 2020, the updated calculator identified 58

of 66 (88%) EOS cases within the first 24 hours after birth, recommending empirical antibiotics in 53 of 66 (80%) cases and blood cultures alone in an additional 8% of cases. Only 32 of 66 (48%) EOS cases had signs of illness at birth or had an EOS risk of ≥ 3 per 1000 live births at birth, emphasizing the importance of good clinical surveillance for all infants in the first 24 hours after birth. No risk stratification approach will identify 100% of EOS cases at birth, for several reasons. First, not every EOS case occurs among infants born in the setting of established clinical risk factors. In this study, 12% of infants with EOS had estimated low sepsis risk based on risk factors but presented with signs of illness after 24 hours of age. EOS cases that present >24 hours after birth likely represent newborns who are not infected at birth but colonized with high-risk pathogens. In such cases, the infant becomes ill with progression from colonization to invasive infection occurring after birth. Second, risk stratification models can only use available risk data. Cases of GBS EOS continue to occur among infants born to pregnant persons who screen negative for GBS.¹²

TABLE 3 Model Comparisons (July 2014 to December 2020 Birth Cohort)

Original Model					Updated Model				
Clinical Classification	EOS Cases Identified	Sensitivity	Empirical Antibiotics	Antibiotic Rate	EOS Cases Identified	Sensitivity	Empirical Antibiotics	Antibiotic Rate	Antibiotics per Additional Case Identified
At birth	30	0.45	4393	1.7%	32	0.48	4408	1.7%	7.5
12 h	43	0.65	8466	3.2%	46	0.70	8737	3.3%	90.3
24 h	50	0.76	9381	3.5%	53	0.80	9854	3.7%	157.7

Population $n = 264\,347$; EOS cases $n = 66$.

TABLE 4 Characteristics of EOS Infants with EOS risk <3 per 1000 Live Births and Not Meeting Criteria for Clinical Illness in the First 24 h in the Original or Updated Model

Case	Symptoms Time (h)	Blood Culture Time ^a (h)	Organism	Repeat Blood Culture Sterile Before Abx	>1 Positive Blood Culture	Highest EOS Risk Per 1000 Births (Original)	Highest EOS Risk Per 1000 Births (Updated)	TTP (h)	Comments
1	Never	4	GBS	Yes	No	2.35	6.33	21	Well-appearing
2	Never	4	GBS	Yes	No	2.83	4.96	44	Well-appearing
3	Never	23	<i>K. pneumoniae</i>	Yes	No	0.07	0.12	24	ABO incompatibility, CBC left shift prompting blood culture,
4	Never	9	<i>E. coli</i>	Yes	No	1.15	1.50	20	Well-appearing
5	Birth	1	<i>E. coli</i>	Yes	No	1.25	1.63	20	Grunting and tachypnea, Equivocal
6	8	14	<i>S. aureus</i>	Yes	No	1.80	2.64	20	Hypothermia, Equivocal
7	9	40	<i>E. coli</i>	Yes	No	2.24	3.30	44	Tachypnea, Equivocal
8	11	14	GBS	No	Yes	1.53	1.96	18	Grunting and tachypnea, Equivocal
9	20	22	<i>E. coli</i>	No	No	1.36	1.64	19	Respiratory distress, Equivocal
10	24	30	<i>E. coli</i>	No	Yes	0.92	1.21	28	Hypothermia, hypoglycemia, and poor feeding, Equivocal, <i>E. coli</i> in CSF
11	28	30	<i>E. coli</i>	No	Yes	1.64	2.10	33	Tachypnea
12	28	32	<i>E. coli</i>	No	No	0.04	0.07	19	Apnea
13	35	36	GBS	No	No	0.11	0.16	18	Fever
14	38	40	<i>E. coli</i>	No	No	0.09	0.16	17	Hypoglycemia, cardiorespiratory collapse, NEC
15	>48	72	<i>S. aureus</i>	No	No	0.05	0.09	21	Poor feeding, lethargy, fever
16	50	53	<i>E. faecalis</i>	No	No	0.02	0.02	25	Irritability, fever, hypoglycemia

Abx, antibiotics; CBC, complete blood count; CSF, cerebrospinal fluid; NEC, necrotizing enterocolitis; TTP, time to positivity (collection time to blood culture positivity).

^a Time after birth blood culture was collected.

Finally, in this study, we identified 2 cases that occurred >48 hours after birth, with pathogens atypical for EOS among term infants. Such cases may have a different pathogenesis of infection that is not captured by the predictors in our models and perhaps represent cases of nosocomial infection. It is also important to note that an accepted practice at our centers is to obtain blood cultures without the initiation of empirical antibiotics at an estimated risk between 1/1000 and 3/1000. This practice allowed us to identify 7 EOS cases in which a repeat blood culture before the initiation of antibiotics was sterile. Such cases likely represent transient bacteremia, and the clinical benefit of treating such cases as invasive infections remains unclear.

Other approaches to risk stratification, such as categorical risk factor assessment^{19,32,33} or assessment based on newborn condition,^{34–36} have been used. The advantage of the KP EOS calculator is the explicit calculation of risk and treatment thresholds, which allows for the evaluation of the trade-off between prompt identification of EOS cases and antibiotic exposure among uninfected infants. Recalibration of model coefficients or different statistical methodologic choices will result in slightly different estimations of EOS risk. These differences only have an impact if they are shifting infants below or above treatment thresholds. The treatment thresholds selected have as much, if not more, impact on the sensitivity and specificity of the model. Reductions in empirical antibiotic administration can be

achieved by raising treatment thresholds, but this requires that clinicians and families accept that more infants will then be identified on the basis of advanced clinical illness. The appropriate choice of treatment thresholds and associated trade-offs may be different for each hospital, each physician, or even each family. Supplemental Table 5 shows the trade-off between the sensitivity of the updated model and antibiotic use using different treatment thresholds. Further reductions in empirical antibiotic use among term infants may also be achieved by consideration of the low-EOS risk phenotype initially described to risk-stratify preterm infants; those born for maternal or fetal noninfection-related reasons, by cesarean delivery in the absence of labor or attempts to induce labor, with rupture of membranes (ROM) at delivery are shown to have a near-zero risk of EOS.^{23,37–39}

A strength of our study is our ability to leverage detailed clinical, physiologic, and microbiologic data for a study cohort of 412 595 infants. A limitation of our study is the use of objective physiologic data in the electronic medical record to determine clinical status, which may have missed infants with subjective signs such as retractions and grunting. Our objective criteria may have also

differed from clinicians' real-time assessments. Low-EOS risk phenotype term infants may not have been evaluated by blood culture nor received empirical antibiotics despite needing respiratory support for ≥ 2 hours because their previous probability of infection was near zero.³⁹

In conclusion, both the original and updated KP EOS calculators are effective risk stratification tools for quantifying EOS risk among infants born at ≥ 35 weeks' gestation.

ABBREVIATIONS

CI: confidence interval
EOS: early-onset sepsis
GA: gestational age
GBS: group B *Streptococcus*
IAP: intrapartum antibiotic prophylaxis
KPNC: Kaiser Permanente Northern California
LR: likelihood ratio

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REFERENCES

- Puopolo KM, Draper D, Wi S, et al. Estimating the probability of neonatal early-onset infection on the basis of maternal risk factors. *Pediatrics*. 2011;128(5):e1155–e1163
- Escobar GJ, Puopolo KM, Wi S, et al. Stratification of risk of early-onset sepsis in newborns ≥ 34 weeks' gestation. *Pediatrics*. 2014;133(1):30–36
- Kuzniewicz MW, Walsh EM, Li S, et al. Development and implementation of an early-onset sepsis calculator to guide antibiotic management in late preterm and term neonates. *Jt Comm J Qual Patient Saf*. 2016;42(5):232–239
- Kuzniewicz MW, Puopolo KM, Fischer A, et al. A quantitative, risk-based approach to the management of neonatal early-onset sepsis. *JAMA Pediatr*. 2017;171(4):365–371
- Achten NB, Dorigo-Zetsma JW, van der Linden PD, et al. Sepsis calculator implementation reduces empiric antibiotics for suspected early-onset sepsis. *Eur J Pediatr*. 2018;177(5):741–746
- Achten NB, Klingenberg C, Benitz WE, et al. Association of use of the neonatal early-onset sepsis calculator with reduction in antibiotic therapy and safety: a systematic review and meta-analysis. *JAMA Pediatr*. 2019;173(11):1032–1040
- Deshmukh M, Mehta S, Patole S. Sepsis calculator for neonatal early onset sepsis – a systematic review and meta-analysis. *J Matern Fetal Neonatal Med*. 2021;34(11):1832–1840
- Dhudasia MB, Mukhopadhyay S, Puopolo KM. Implementation of the sepsis risk calculator at an academic birth hospital. *Hosp Pediatr*. 2018;8(5):243–250
- Strunk T, Buchiboyina A, Sharp M, et al. Implementation of the neonatal sepsis calculator in an Australian tertiary perinatal centre. *Neonatology*. 2018;113(4):379–382
- Puopolo KM, Benitz WE, Zaoutis TE; Committee on Fetus and Newborn; Committee on Infectious Diseases. Management of neonates born at ≥ 35 0/7 weeks' gestation with suspected or proven early-onset bacterial sepsis. *Pediatrics*. 2018;142(6):e20182894
- Prevention of group b streptococcal early-onset disease in newborns: ACOG Committee Opinion, Number 797. *Obstet Gynecol*. 2020;135(2):e51–e72

12. Nanduri SA, Petit S, Smelser C, et al. Epidemiology of invasive early-onset and late-onset group b streptococcal disease in the United States, 2006 to 2015: multistate laboratory and population-based surveillance. *JAMA Pediatr.* 2019;173(3): 224–233
13. Castor ML, Whitney CG, Como-Sabetti K, et al. Antibiotic resistance patterns in invasive group B streptococcal isolates. *Infect Dis Obstet Gynecol.* 2008;2008:727505
14. Teatero S, Ferrieri P, Martin I, et al. Serotype distribution, population structure, and antimicrobial resistance of group b streptococcus strains recovered from colonized pregnant women. *J Clin Microbiol.* 2017;55(2):412–422
15. Bulska M, Szcześniak P, Pięta-Dolińska A, et al. The placental transfer of erythromycin in human pregnancies with group B streptococcal infection. *Ginekol Pol.* 2015;86(1):33–39
16. Benitz WE, Achten NB. Technical assessment of the neonatal early-onset sepsis risk calculator. *Lancet Infect Dis.* 2021;21(5): e134–e140
17. Kuzniewicz MW, Escobar GJ, Newman TB. Impact of universal bilirubin screening on severe hyperbilirubinemia and phototherapy use. *Pediatrics.* 2009;124(4):1031–1039
18. Chimmula S, Dhuru R, Folck B, et al. PS2-44: VDW data sources: Kaiser Permanente Northern California. *Clin Med Res.* 2012; 10(3):193
19. Schrag S, Gorwitz R, Fultz-Butts K, Schuchat A. Prevention of perinatal group B streptococcal disease. Revised guidelines from CDC. *MMWR Recomm Rep.* 2002;51(RR-11):1–22
20. Creti R, Imperi M, Berardi A, et al; Italian Neonatal GBS Infections Working Group. Neonatal group b streptococcus infections: prevention strategies, clinical and microbiologic characteristics in 7 years of surveillance. *Pediatr Infect Dis J.* 2017;36(3):256–262
21. Nanovskaya T, Patrikeeva S, Zhan Y, et al. Transplacental transfer of vancomycin and telavancin. *Am J Obstet Gynecol.* 2012;207(4): e331–e336
22. Hnat MD, Gainer J, Bawdon RE, Wendel GD Jr. Transplacental passage of vancomycin in the ex vivo human perfusion model. *Infect Dis Obstet Gynecol.* 2004;12(2):57–61
23. Puopolo KM, Lynfield R, Cummings JJ; Committee on Fetus and Newborn; Committee on Infectious Diseases. Management of infants at risk for group b streptococcal disease. *Pediatrics.* 2019;144(2):e20191881
24. Colombo DF, Lew JL, Pedersen CA, et al. Optimal timing of ampicillin administration to pregnant women for establishing bactericidal levels in the prophylaxis of group B streptococcus. *Am J Obstet Gynecol.* 2006;194(2):466–470
25. de Cueto M, Sanchez MJ, Sampedro A, et al. Timing of intrapartum ampicillin and prevention of vertical transmission of group B streptococcus. *Obstet Gynecol.* 1998;91(1):112–114
26. Lin FY, Brenner RA, Johnson YR, et al. The effectiveness of risk-based intrapartum chemoprophylaxis for the prevention of early-onset neonatal group B streptococcal disease. *Am J Obstet Gynecol.* 2001;184(6):1204–1210
27. Illuzzi JL, Bracken MB. Duration of intrapartum prophylaxis for neonatal group B streptococcal disease: a systematic review. *Obstet Gynecol.* 2006;108(5):1254–1265
28. Gengsheng Qin, Hotilovac L. Comparison of non-parametric confidence intervals for the area under the ROC curve of a continuous-scale diagnostic test. *Stat Methods Med Res.* 2008;17(2):207–221
29. Carpenter J, Bithell J. Bootstrap confidence intervals: when, which, what? A practical guide for medical statisticians. *Stat Med.* 2000;19(9):1141–1164
30. Austin PC, Steyerberg EW. The Integrated Calibration Index (ICI) and related metrics for quantifying the calibration of logistic regression models. *Stat Med.* 2019;38(21):4051–4065
31. King S, Zeng L. Logistic regression in rare events data. *Polit Anal.* 2001;9:137–163
32. Centers for Disease Control and Prevention. Prevention of perinatal group B streptococcal disease: a public health perspective. *MMWR Recomm Rep.* 1996;45(RR-7):1–24
33. Schuchat A, Zywicki SS, Dinsmoor MJ, et al. Risk factors and opportunities for prevention of early-onset neonatal sepsis: a multicenter case-control study. *Pediatrics.* 2000;105(1 Pt 1):21–26
34. Cantoni L, Ronfani L, Da Rioli R, Demarini S; Perinatal Study Group of the Region Friuli-Venezia Giulia. Physical examination instead of laboratory tests for most infants born to mothers colonized with group B streptococcus: support for the Centers for Disease Control and Prevention's 2010 recommendations. *J Pediatr.* 2013;163(2):568–573
35. Berardi A, Fornaciari S, Rossi C, et al. Safety of physical examination alone for managing well-appearing neonates ≥ 35 weeks' gestation at risk for early-onset sepsis. *J Matern Fetal Neonatal Med.* 2015;28(10):1123–1127
36. Joshi NS, Gupta A, Allan JM, et al. Clinical monitoring of well-appearing infants born to mothers with chorioamnionitis. *Pediatrics.* 2018;141(4):e20172056
37. Coggins SA, Mukhopadhyay S, Triebwasser J, et al. Association of delivery risk phenotype with early-onset sepsis in preterm infants. *J Perinatol.* 2023;43(9):1166–1172
38. Puopolo KM, Mukhopadhyay S, Hansen NI, et al; NICHD Neonatal Research Network. Identification of extremely premature infants at low risk for early-onset sepsis. *Pediatrics.* 2017;140(5):e20170925
39. Flannery DD, Mukhopadhyay S, Morales KH, et al. Delivery characteristics and the risk of early-onset neonatal sepsis. *Pediatrics.* 2022;149(2):e2021052900
40. Hosmer D, Lemeshow S. *Applied Logistic Regression.* 2nd ed. New York, NY: John Wiley and Sons; 2000